

EDUCATION

•Fudan University

Bachelor in Software Engineering. Advisor: Prof. Bihuan Chen

September 2019 - June 2024

Shanghai, China

•The Chinese University of Hong Kong, Shenzhen

PhD in Computer Science. Advisor: Prof. Zhizheng Wu

Research Topics: Audio Enhancement, Music Generation.

September 2024 - Present

Shenzhen, China

SELECTED PUBLICATIONS

Full list at Google Scholar, *Equal Contribution.

- **Junan Zhang**, Jing Yang, Zihao Fang, Yuancheng Wang, Zehua Zhang, Zhuo Wang, Fan Fan, Zhizheng Wu. ANYENHANCE: A Unified Generative Model with Prompt-Guidance and Self-Critic for Voice Enhancement *Accepted. IEEE Transactions on Audio, Speech, and Language Processing (TASLP)*, 2025.
- **Junan Zhang**, Chong Jing, Jing Yang, Yingda Shen, Yuancheng Wang, Fan Fan, Zhizheng Wu. ANYENHANCE2: Controlling Generative Freedom for Faithful and Universal Audio Restoration *Technical Report*, 2026.
- **Junan Zhang***, Yunjia Zhang*, Xueyao Zhang, Zhizheng Wu. ANYACCOMP: Generalizable Accompaniment Generation via Quantized Melodic Bottleneck *Accepted. IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP)*, 2026.
- Yunjia Zhang*, **Junan Zhang***, Jing Yang*, Xueyao Zhang, Fan Fan, Zhizheng Wu. ANYEDIT: A Unified Framework for Speech and Singing Voice Editing with Real-World Environmental Consistency *Accepted. Annual Conference on Neural Information Processing Systems (NeurIPS)*, 2026.
- **Junan Zhang**, Xueyao Zhang, Jing Yang, Yuancheng Wang, Fan Fan, Zhizheng Wu. Multi-Metric Preference Alignment for Generative Speech Restoration *Accepted. AAAI Conference on Artificial Intelligence (AAAI)*, 2026.
- Xueyao Zhang, **Junan Zhang**, Yuancheng Wang, Chaoren Wang, Yuanzhe Chen, Dongya Jia, Zhuo Chen, Zhizheng Wu. Vevo2: A Unified and Controllable Framework for Speech and Singing Voice Generation *Accepted. IEEE Transactions on Audio, Speech, and Language Processing (TASLP)*, 2025.
- Yicheng Gu, **Junan Zhang**, Chaoren Wang, Jerry Li, Zhizheng Wu, Lauri Juvela. Aliasing Free Neural Audio Synthesis *Accepted. IEEE Transactions on Audio, Speech, and Language Processing (TASLP)*, 2026.
- Yuancheng Wang, Jiachen Zheng, **Junan Zhang**, Xueyao Zhang, Huan Liao, Zhizheng Wu. Metis: A Foundation Speech Generation Model with Masked Generative Pre-training *Accepted. Annual Conference on Neural Information Processing Systems (NeurIPS)*, 2025.
- **Junan Zhang**, Mengyao Zhu, Xin Xu, Hui Bu, Zhenhua Ling, Zhizheng Wu. The CCF AATC 2025 Speech Restoration Challenge: A Retrospective *Technical Report*, 2025.
- Junyan Ye*, Baichuan Zhou*, Zilong Huang*, **Junan Zhang***, Tianyi Bai, Hengrui Kang, Jun He, Honglin Lin, Zihao Wang, Tong Wu, Zhizheng Wu, Yiping Chen, Dahua Lin, Conghui He, Weijia Li. LOKI: A Comprehensive Synthetic Data Detection Benchmark using Large Multimodal Models *Accepted (Spotlight). International Conference on Learning Representations (ICLR)*, 2025.
- **Junan Zhang**, Kaifeng Huang, Yiheng Huang, Bihuan Chen, Ruisi Wang, Chong Wang, Xin Peng. Malicious Package Detection in NPM and PyPI using a Single Model of Malicious Behavior Sequence *Accepted. ACM Transactions on Software Engineering and Methodology (TOSEM)*, 2024.

RESEARCH EXPERIENCE

•ByteDance Seed

December 2025 - Present

Research Scientist Intern (Music-to-Music Generation)

Shenzhen, China

- Participated in the development of the next-generation SeedMusic model, focusing on the algorithm implementation of the music-to-music (m2m) generation.
- Implemented core music-to-music functionalities, including cover song generation, timbre cloning, timbre-cloned cover generation, reference-based generation, mashup, and motif-guided generation.
- Developed post-training (reinforcement learning) pipelines for timbre-cloned cover generation to improve audio quality and timbre similarity.

•CCF Advanced Audio Technology Competition 2025

June 2025 - August 2025

Lead Organizer (Track 1: Speech Restoration)

Shenzhen, China

- Spearheaded the complete technical design and execution of the Speech Restoration track for CCF Advanced Audio Technology Competition (AATC 2025) hosted by the China Computer Federation (CCF).
- Engineered and open-sourced the official baseline system based on the ANYENHANCE model, providing a robust and competitive starting point for participating teams.
- Defined the competition's entire technical architecture, from problem formulation and data curation to the comprehensive evaluation protocol (e.g. DNSMOS, PESQ, complexity constraints), attracting over 100 registered teams.

•Shanghai AI Lab

January 2024 - June 2024

Undergraduate Research Intern

Shanghai, China

- Co-developed a comprehensive benchmark for detecting synthetic data via large multimodal models.
- Conducted extensive research on synthetic datasets and open/closed-source models across four domains: speech, singing, audio, and music. Collected and generated 2,280 real/synthetic parallel audios, curated corresponding QA datasets for benchmarking.
- Evaluated mainstream audio multimodal large models, including Qwen-Audio, SALMONN, AnyGPT, and Gemini-1.5-Flash. Discovered that current audio multimodal large models struggle to distinguish synthetic audio from real audio (all ~50% accuracy), with this benchmark forming the foundation of the LOKI project, accepted by ICLR 2025 (Spotlight).

•Software Engineering Lab, Fudan University

May 2022 - June 2023

Undergraduate Research Assistant

Shanghai, China

- Led a project leveraging pre-trained language models (BERT, RoBERTa) and malicious behavior sequence modeling to detect malicious packages in PyPI/NPM, outperforming state-of-the-art methods by 10.8% in precision and 7.3% in recall.
- Discovered 1,482 novel malicious packages in the wild (683 PyPI, 799 NPM), leading to 707 official acknowledgements from the PyPI and NPM security teams.
- Published the complete methodology and findings as a first-author paper in TOSEM 2024.

PROJECT EXPERIENCE

•Amphion Development

November 2023 - Present

Core Contributor

- Acted as a core contributor to Amphion, an audio generation toolkit with over **10k** GitHub stars.
- Led the development and integration of multiple key modules, including Speech Enhancement (SE), Singing Voice Conversion (SVC), and Text-to-Music (TTM) generation.
- Contributed two original, novel models to the framework: ANYENHANCE for unified voice enhancement and ANYACCOMP for accompaniment generation.
- Implemented and integrated a VITS-based model for singing voice conversion.

HONORS & AWARDS

•Best Presentation Award, NCMMSC 2025

October 2025

•Merit Fellowship for PhD Students, The Chinese University of Hong Kong, Shenzhen

September 2024

•Scholarship for Outstanding Students, Fudan University (Top 20%)

November 2023, November 2022

SKILLS

Programming Languages: Python, Bash, LaTeX

Developer Tools: Linux, Git

Framework: PyTorch, Hugging Face, Transformers

Language: Chinese(Native), English(Fluent, TOEFL 102(S 23))

Music Production: Semi-Professional, over 2,000,000 streams on Netease Cloud Music